Teaching demonstration

Understanding Inflation

Why Prices Rise and How Central Banks Respond

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Three questions for today

1. What is inflation and how is it measured?
2. Is inflation good or bad?
3. What do central banks do, and at what cost?

A receipt from the same store, 24 years apart



Item	2002	2026	Change
Rice, 5 kg	¥980	¥4,200	+329%
Milk, 1 L	¥98	¥260	+165%
Eggs, 10	¥78	¥310	+297%
Bread, 1 loaf	¥68	¥210	+209%
Cabbage	¥58	¥230	+297%
Total	**¥1,282**	**¥5,210**	**+306%**

Store: JUSCO Hadano, 12 May 2002 → AEON Hadano, 12 Oct 2026

(same supermarket; JUSCO was later renamed AEON)

Average of the five product increases: $(329 + 165 + 297 + 209 + 297) / 5 = \mathbf{259\%}$

Average inflation per year (compound): $\left(\frac{\text{final}}{\text{initial}}\right)^{1/\text{years}} - 1 = \left(\frac{5210}{1282}\right)^{1/24} - 1 \approx \mathbf{6\%}$

From the receipt to the inflation rate

Do you think 6% is the inflation rate?

Milk rose 165%; rice rose 329%.
Whose basket — and how should items be weighted?

The Consumer Price Index

1. Definition

The cost of a fixed basket of goods and services, relative to a base period.

Basket from household spending surveys

(average purchases in the base year).

2. Weights

Weight = category spending ÷ total spending.

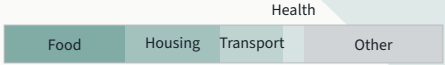
Larger share of spending ⇒ larger weight.

3. Formula

$$\mathrm{CPI}_t = 100 \times \frac{\sum_j p_{tj}\, q_{0j}}{\sum_j p_{0j}\, q_{0j}}$$

j = item · p_{tj} = today's price · q_{0j} = base quantity

What's in the basket



Food	27.5%
Housing	21.8%
Transport	14.4%
Health	4.7%
Other	31.5%

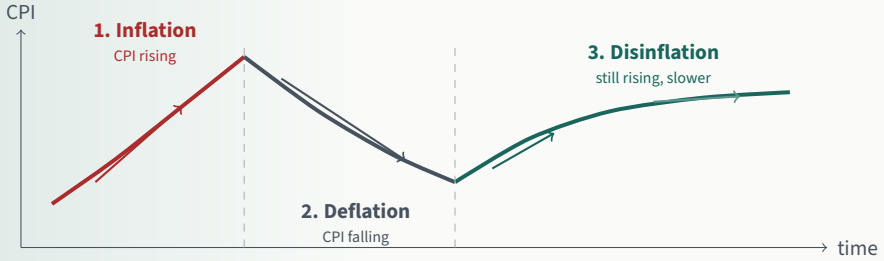
Statistics Bureau of Japan, 2025 base

Inflation, Deflation, and Disinflation

CPI = a *level*. The inflation rate measures how fast that level is changing:

$$\pi_t = \frac{\text{CPI}_t - \text{CPI}_{t-1}}{\text{CPI}_{t-1}}$$

1. Inflation: CPI rising ($\pi > 0$). **2. Deflation:** CPI falling ($\pi < 0$). **3. Disinflation:** CPI still rising, but π falling.



Example: milk and rice

Question

Basket $=$ **2 L milk** $+$ **2 bags rice**. Compute the CPI at $t = 0, 1, 2$, then π_1 and π_2. Is $t = 2$ deflation?

Prices

	$t = 0$ (base)	$t = 1$	$t = 2$
Milk	¥100	¥120	¥125
Rice	¥200	¥240	¥250

Example: milk and rice

Question

Basket = **2 L milk** + **2 bags rice**. Compute the CPI at $t = 0, 1, 2$, then π_1 and π_2. Is $t = 2$ deflation?

Prices

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Solution

	$t = 0$	$t = 1$	$t = 2$
Cost $= 2p_m + 2p_r$	$2 \times 100 + 2 \times 200 =$ **600**	$2 \times 120 + 2 \times 240 =$ **720**	$2 \times 125 + 2 \times 250 =$ **750**
CPI $= 100 \times$ cost$/600$	**100**	**120**	**125**
π_t	—	$(120 - 100)/100 =$ **20%**	$(125 - 120)/120 \approx$ **4.2%**

CPI rose both periods $\Rightarrow$ **inflation**. π fell 20% $\rightarrow$ 4.2% $\Rightarrow$ **disinflation**, not deflation.